Graph Partitions and Gerrymandering-Resistant Redistricting in Hong Kong

Mingkai Liu

January 7, 2026

Abstract

We model Hong Kong's 44 District Council geographical constituencies as connected partitions of 452 historical constituency areas. A fixed-target Monte Carlo analysis generates 96,000 recorded maps under three local population bands and compares the authenticated official map with those alternatives. We then design Plan F using a group-label-invariant objective that minimizes partisan seat–vote asymmetry under explicit historical voting scenarios. Official 2019 vote totals and ballot-endorsement records provide the electoral inputs; unclassified votes remain uncertain. The model respects the two-seat, single-vote electoral rule and varies both unclassified-vote allocation and within-group candidate splitting. Plan F has zero equal-support seat bias in the original nine scenarios and a maximum seat–vote symmetry defect of 1 seat on the expanded 225-point design grid, compared with 7 for the official map. It does not pass the stronger all-25 zero-bias test. A denser post-design grid reveals a maximum equal-support bias of 4 seats, limiting any gerrymandering-free designation to the stated nine-scenario criterion. Plan F retains all 18 administrative districts, assigns every source area exactly once, preserves specified island and rural units, and passes independent graph and population checks. A conditional definition of "gerrymandering-free" separates a verifiable finite-model certificate from an unsupported claim about all political preferences or elections. The proposal and its maps demonstrate what boundary design can improve, while a variance decomposition shows why redistricting alone cannot remove inequality caused by constituency allocation across administrative districts. Reproducible assignments, source snapshots, simulation records, and audit programs accompany the paper.

Keywords: Hong Kong; graph partitions; Monte Carlo; gerrymandering; partisan symmetry; two-seat elections.
Data and revision cutoff: October 13, 2026. The cover date is retained from the research record; the computations reported here include subsequent revisions.

1 Research question and institutional setting

This paper asks whether Hong Kong's constituency boundaries can be evaluated against a reproducible collection of alternatives, and whether a complete replacement map can improve both population balance and a specified measure of partisan fairness. The empirical object is the 44 District Council geographical constituencies introduced for the 2023 election. It is not a study of Legislative Council boundaries, nor a claim about every form of representation in Hong Kong.

The work has three linked components. First, we authenticate the official map and represent it as a partition of a population-weighted adjacency graph. Second, we generate Monte Carlo ensembles under declared geographical and population constraints and locate the official map

within those distributions. Third, we construct Plan F, a new graph partition chosen by an objective that treats the two modeled political groups symmetrically. The final component adds voting scenarios to a problem that population counts and polygon geometry alone cannot resolve.

1.1 What the 44 constituencies represent

The 2023 District Council reform retained the 18 administrative districts and introduced 44 geographical constituencies returning 88 directly elected members. Each elector votes for one candidate; the two candidates with the highest vote totals in the constituency are elected. The directly elected geographical seats are only one component of the District Councils, which also contain appointed, indirectly elected, and ex-officio members [11]. Consequently, changing these 44 boundaries does not redesign the entire institution.

The official election report describes an exceptional 2023 delineation process in which the government merged existing District Council constituency areas into larger geographical constituencies. It sought broadly similar populations within the same administrative district and preservation of community integrity [1, paras. 2.12–2.17]. We reconstruct this actual official assignment, rather than treating a randomly generated seed map as the observed plan.

The district-specific population quota follows the member allocation. If district d has population P_d, returns B_d geographical members in total, and constituency i returns m_i members, its relevant population number is

$$q_{di} = \frac{P_d}{B_d} m_i. \tag{1.1}$$

Here $B_d = 2k_d$ and $m_i = 2$, where k_d is the district's number of geographical constituencies, so the quota reduces to $q_d = P_d/k_d$ [2]. A comparison against one territory-wide quota answers a different question from a comparison against these local quotas.

1.2 The meaning of a legal-map ensemble

An ensemble of *legal maps* would need to encode all applicable requirements, using the population basis required for the relevant boundary review. Connectivity and population equality are essential computational constraints but are not the whole review. Community identity, local ties, physical geography, building assignments, and practical access may require evidence outside a polygon adjacency graph.

Our ensembles are therefore described as *constraint-admissible graph maps*. They retain administrative-district nesting and constituency counts, keep every source area intact, and satisfy specified local population bands. The 25% band is motivated by the ordinary-review tolerance discussed by the EAC, while 10% and 15% are stricter comparison scenarios. The available 2027 announcement is provisional and uses projected populations; our empirical snapshot uses 2021 census counts [4]. Passing our 25% test does not certify a recommendation as legally approved or compliant on a different population date.

This distinction limits the language of the contribution, but not its computational substance. We deliver actual assignments, an observed-map reconstruction, fixed-target chains, independent feasibility checks, and a complete proposal. The remaining legal and community checks are identified so that another researcher or boundary-review body can extend the model.

1.3 How an ensemble can investigate gerrymandering

Gerrymandering concerns the use of boundaries to advantage a political group. An unusually shaped region or an unequal population is not sufficient evidence of that advantage. Conversely, equal populations and short boundaries do not exclude politically consequential packing or fragmentation of supporters. An electoral model must attach voting information to the geographical units and specify how those votes become seats.

Herschlag and colleagues demonstrate a comparison of historical voting outcomes across Monte Carlo redistricting ensembles [14]. That framework motivates the extension here, but Hong Kong's two-seat rule requires a different seat-conversion model. We cannot import a one-seat majority formula or an American two-party efficiency-gap threshold and describe the resulting number as a validated Hong Kong fairness verdict.

Our proposed solution uses a transparent historical endorsement grouping, explicit uncertainty scenarios, and a two-seat candidate-splitting model. It tests the official map and Plan F on the same inputs. The term "gerrymandering-free" is reserved for a stated, conditional graph-partition criterion in Section 4; an unqualified political certificate is not asserted.

2 Constituencies as graph partitions

2.1 Data and authenticated baseline

The atomic geography consists of the 452 official 2019 District Council constituency areas (DCCAs). Their official digital polygons are joined by source-area code to the 2021 census population counts [6, 7]. The populations sum to 7,411,945. We overlay those polygons with the official 2023 constituency geometry, restrict matches to the same administrative district, and assign each atom to the constituency containing the greatest proportion of its area [5, 3].

This crosswalk recovers all 44 published constituency population totals exactly. Each atom has at least 99.9997904% of its projected area inside its assigned constituency; none falls below the stated 99.99% match threshold. Thus the observed comparator is an authenticated reconstruction of the official map on the same units used for every proposed and sampled map. The public reproduction archive contains the complete crosswalk, population tables, input hashes, and containment audit.

We transform geometry to Hong Kong's projected coordinate system, EPSG:2326, and normalize coordinates to a one-millimetre precision grid. The normalization avoids missed shared borders caused by floating-point discrepancies. It changes any atom's area by at most 0.000296497% in the symmetric-difference calculation. Sensitivity checks at one millimetre, ten centimetres, and one metre give the same edge set. No nearest-neighbour link, straight-line bridge, or conjectured ferry connection is added.

2.2 The feasible set

Let $G = (V, E)$ be the graph, with $|V| = 452$. Vertex u carries census population p_u, administrative-district code $d(u)$, and a source polygon. An edge joins two vertices when their same-district electoral polygons share a boundary longer than one metre. The resulting graph has 1,060 edges. A map is a function

$$\pi : V \longrightarrow \mathcal{C}, \qquad |\mathcal{C}| = 44, \tag{2.1}$$

where the labels in $\mathcal{C}_d$ denote the k_d constituencies allocated to district d. Set $V_c(\pi) = \{u : \pi(u) = c\}$ and $P_c(\pi) = \sum_{u \in V_c(\pi)} p_u$.

For a local tolerance ϵ, the declared feasible set is

$$\Omega_\epsilon = \{\pi : \pi(u) \in \mathcal{C}_{d(u)}, \quad V_c(\pi) \neq \varnothing, \\ G[V_c(\pi)] \text{ is connected}, \quad |P_c(\pi)/q_{d(c)} - 1| \leq \epsilon \text{ for every } c\}. \tag{2.2}$$

Every vertex is assigned exactly once. The constraints preserve all 18 district boundaries and the official number of constituencies in each district. Labels are bookkeeping devices: they do not create a preference for a named constituency. The official map and the earlier population-oriented Plan P belong to all three supports, $\epsilon \in \{.10, .15, .25\}$.

The electoral polygons contain marine areas. Their total area is approximately 2,754 square kilometres, rather than a land-only area. Consequently, connectedness in G is a reproducible electoral-polygon property, not proof of walking, road, or ferry accessibility. Likewise, a cut-length statistic built from these polygons includes water boundaries. These limitations remain relevant even when all connectivity checks pass.

2.3 Population and boundary statistics

The local population statistic is the root-mean-square deviation from district-specific quotas:

$$L(\pi) = \left[\frac{1}{44}\sum_{c \in \mathcal{C}} \left(\frac{P_c(\pi)}{q_{d(c)}} - 1\right)^2\right]^{1/2}. \tag{2.3}$$

For the territory-wide quota $q = 7,411,945/44$, define

$$C(\pi) = \frac{\left[44^{-1}\sum_c (P_c(\pi) - q)^2\right]^{1/2}}{q}. \tag{2.4}$$

We report $100L$ and $100C$ as percentages. A smaller value means less dispersion according to its specified denominator; it is not itself an electoral-bias score.

Internal cut length is

$$K(\pi) = \sum_{\{u,w\} \in E} \ell_{uw} \mathbf{1}\{\pi(u) \neq \pi(w)\}, \tag{2.5}$$

where ℓ_{uw} is shared-boundary length. We also measure the population reassigned relative to the official map after optimal constituency-label matching within each administrative district. Because no district has more than four constituencies, this matching can be solved by enumerating at most 4! label permutations. It prevents a relabelled copy of the same map from being counted as a disruptive redesign.

2.4 An allocation constraint that redistricting cannot remove

Let $\mu_d = P_d/k_d$ be a district's mean constituency population. An exact decomposition gives

$$\frac{1}{44}\sum_d \sum_{c \in \mathcal{C}_d} (P_c - q)^2 = \underbrace{\frac{1}{44}\sum_d k_d(\mu_d - q)^2}_{V_B} + \underbrace{\frac{1}{44}\sum_d \sum_{c \in \mathcal{C}_d} (P_c - \mu_d)^2}_{V_W}. \tag{2.6}$$

To verify it, expand $P_c - q = (P_c - \mu_d) + (\mu_d - q)$. The cross term vanishes because $\sum_{c \in \mathcal{C}_d}(P_c - \mu_d) = 0$. Internal boundary changes alter V_W but leave V_B fixed.

In the official map, V_B accounts for 89.02% of the global population variance. Even exact equality within each district would leave a territory-wide CV of 12.3909%, compared with the observed 13.1332%. This is an algebraic lower bound, not a claim that a connected whole-DCCA map realizes every equal-division target. It establishes why a geographically feasible boundary proposal can improve local equality without solving territory-wide malapportionment.

3 Monte Carlo ensembles and the official map

3.1 A fixed reference distribution

For each $\epsilon \in \{.10, .15, .25\}$, the target is the uniform law on the declared feasible labelled partitions:

$$\nu_\epsilon(\pi) = \frac{\mathbf{1}\{\pi \in \Omega_\epsilon\}}{|\Omega_\epsilon|}. \tag{3.1}$$

No electoral data, continuity penalty, compactness energy, or frozen community units enter this reference law. Those omissions are choices, not a claim that such considerations are irrelevant to actual boundary reviews. The reference asks what connected, whole-unit, locally balanced partitions look like when all admissible assignments have equal weight.

Each unlabelled partition has exactly $\prod_d k_d!$ labellings because its nonempty regions can be permuted independently within districts. There are no label-specific constraints. The labelled uniform law therefore induces a uniform law on unlabelled partitions as well. A spanning-tree-weighted law would answer another reference question; the name of a sampling algorithm alone does not identify its target [8, 9].

3.2 The lazy flip–swap kernel

At each clock step, choose uniformly one of the 16 administrative districts with more than one constituency. Within that district:

1. With probability 0.50, retain the state.
2. With probability 0.35, choose a vertex uniformly and choose another constituency label uniformly; propose transferring the vertex to that label.
3. With probability 0.15, choose two distinct vertices uniformly and propose exchanging their labels.

Accept the proposal only when the resulting assignment belongs to Ω_ϵ. A rejected proposal remains a recorded clock state. We do not condition the reference sample on successful moves.

Proposition 3.1 (Reversibility and aperiodicity)**.** *The specified kernel is reversible for ν_ϵ. Every feasible state has a self-transition probability of at least $1/2$.*

Proof. For a flip, the forward and reverse proposals choose the same district and vertex and one of the same $k_d - 1$ alternative labels, so their proposal probabilities agree. A swap reverses itself; either ordered choice of the pair has the same probability in both directions. Feasibility rejection affects only self-transition probability. Hence the off-diagonal transition probabilities satisfy $P(\pi, \pi') = P(\pi', \pi)$, implying detailed balance for a uniform target. The explicit lazy step gives a positive self-loop and therefore aperiodicity on every communicating class. □

This argument does not prove irreducibility on the full Hong Kong feasible set. A chain can be reversible yet fail to reach some admissible configurations. We use multiple starts and

statistic-specific diagnostics, report the remaining limitation, and avoid interpreting empirical ranks as calibrated significance levels.

3.3 Run design and validation

Each band has four random streams with seeds 20260107–20260110. Initial states are the official map, Plan P, a population-focused alternative, and the official map again: there are three starting designs, not four independently generated designs. Each chain discards 1,000,000 warm-up steps, runs 8,000,000 retained clock steps, and records every 1,000th state. Thus each band contains $4 \times 8,000 = 32,000$ records, and the three bands contain 96,000 records. Including warm-up, the computation uses 108,000,000 clock steps.

These are correlated map records and can include repeated assignments. They are not 96,000 independent observations or a complete enumeration. Earlier, shorter pilot runs showed slower movement diagnostics; the common final run length was extended before results were reported. Full assignments, metrics, rejection counts, start states, seeds, software versions, and source hashes accompany the paper.

A separate audit checks all 96,000 records using explicit-set breadth-first traversal rather than the sampler's bit-mask connectivity routine. Every record assigns all 452 atoms once, forms 44 nonempty connected regions, retains district nesting, conserves population, and satisfies its population band. The audit independently recomputes population RMS and CV. A further vectorized check independently reproduces every saved internal cut length to numerical precision.

On a small 3×4 synthetic grid, we enumerate 19 feasible states and build the exact transition matrix. It is symmetric, row-stochastic, and connected. After 200,000 warm-up and 2,000,000 retained clock states, empirical total variation from exact uniform is 0.01415. This validates the implementation on an enumerable problem; it is not evidence that the much larger Hong Kong support has been exhaustively explored.

Rank-normalized and folded split-$\widehat{R}$ diagnostics follow Vehtari and colleagues [10]. All reported aggregate population, cut-length, and movement diagnostics are below 1.01. Effective sizes differ substantially by statistic: at the 10% band, local population RMS has bulk ESS about 18,849, cut length about 2,148, and matched movement about 865. A single nominal map count would conceal this difference. The complete diagnostics are supplied with the records rather than repeated in a long appendix.

3.4 What the simulations show

The official map has local RMS 4.385%. Under the 10% reference, 805 of 32,000 records have RMS at or below that value, an empirical fraction of 2.516%. The four chain fractions range from 2.400% to 2.588%. Thus the official map lies toward the better-balanced end of this reference distribution. It would be incorrect to use these results to describe its local population balance as unusually poor.

Mean local RMS is 5.37%, 8.09%, and 13.34% for the 10%, 15%, and 25% bands respectively. Wider population bands admit more loosely balanced configurations, so changing the percentile can reflect a changed reference question. No records in the two wider bands have local RMS as small as the official map's. Empty observed tails do not imply zero target probability; no independent-binomial confidence bound is applied to correlated chain states.

The official internal cut length is 213.15 kilometres, below every recorded alternative. Ensemble

means are about 367 kilometres. This describes simplicity under the stated polygon statistic, not transport access or political neutrality. The graphs in Figure 1 display both quantities to keep population and boundary complexity separate from the partisan analysis that follows.

4 Gerrymandering-free graph partitions: a conditional criterion

The geographical graph supports an electoral-bias calculation only after votes and an election rule have been specified. This section defines a falsifiable finite-model criterion for a proposed graph partition, rather than treating "gerrymandering-free" as a synonym for compact or population-balanced. A passing map is described as *gerrymandering-free under the equal-support criterion*. That statement is conditional on the group definitions, scenario family, electoral model, and finite test grid given below.

4.1 Official historical inputs and unclassified votes

We extract official 2019 candidate-level results for all 452 DCCAs, comprising 1,090 candidates and 2,931,766 valid votes. The snapshot includes the corrected result recorded on the official results page. The election-result table supplies candidate number, constituency code, name, vote count, and elected status [12]. Only the first two fields are used as join keys; elected status is not used to infer a losing candidate's political affiliation.

The official lists of ballot names and emblems record prescribed-body endorsements requested by candidates and consented to by those bodies [13]. We join every result to its ballot-record row. For an explicit endorsement-based comparison, group A contains candidates associated in those rows with Power for Democracy, the Democratic Party, Civic Party, ADPL, Neo Democrats, Labour Party, League of Social Democrats, or People Power. Group B contains recorded DAB, FTU, New People's Party, BPA, Liberal Party, Kowloon West New Dynamic, Federation of Hong Kong and Kowloon Labour Unions, or New Territories Association of Societies endorsements. Multiple endorsements within a group are counted once. There are no rows matching both defined groups.

These are defined endorsement groups, not a complete reconstruction of two political camps. Group A has 1,153,279 votes from 318 candidates; group B has 790,401 votes from 288 candidates. The remaining 988,086 votes, from 484 candidates, stay unclassified. Known endorsements account for 66.297% of the votes. We do not silently identify an unendorsed independent with either group, and do not infer preferences from census demographics.

The source archive retains each raw ballot row, its source file, any matched bodies, and the assigned code A, B, or U. A reviewer can therefore change the grouping without re-extracting the voting records. Ballot endorsement is still a proxy for an electoral support group, and transferring historical votes to a counterfactual election changes the choice set available to voters. Those assumptions must remain visible.

4.2 Historical support scenarios

For source area u, let A_u, B_u, U_u be the three vote totals and $t_u = A_u + B_u + U_u$. A scenario parameter r allocates a fraction of the unclassified votes to the modeled group A:

$$a_u(r) = \frac{A_u + rU_u}{t_u}, \qquad r \in \{0, .25, .50, .75, 1\}. \tag{4.1}$$

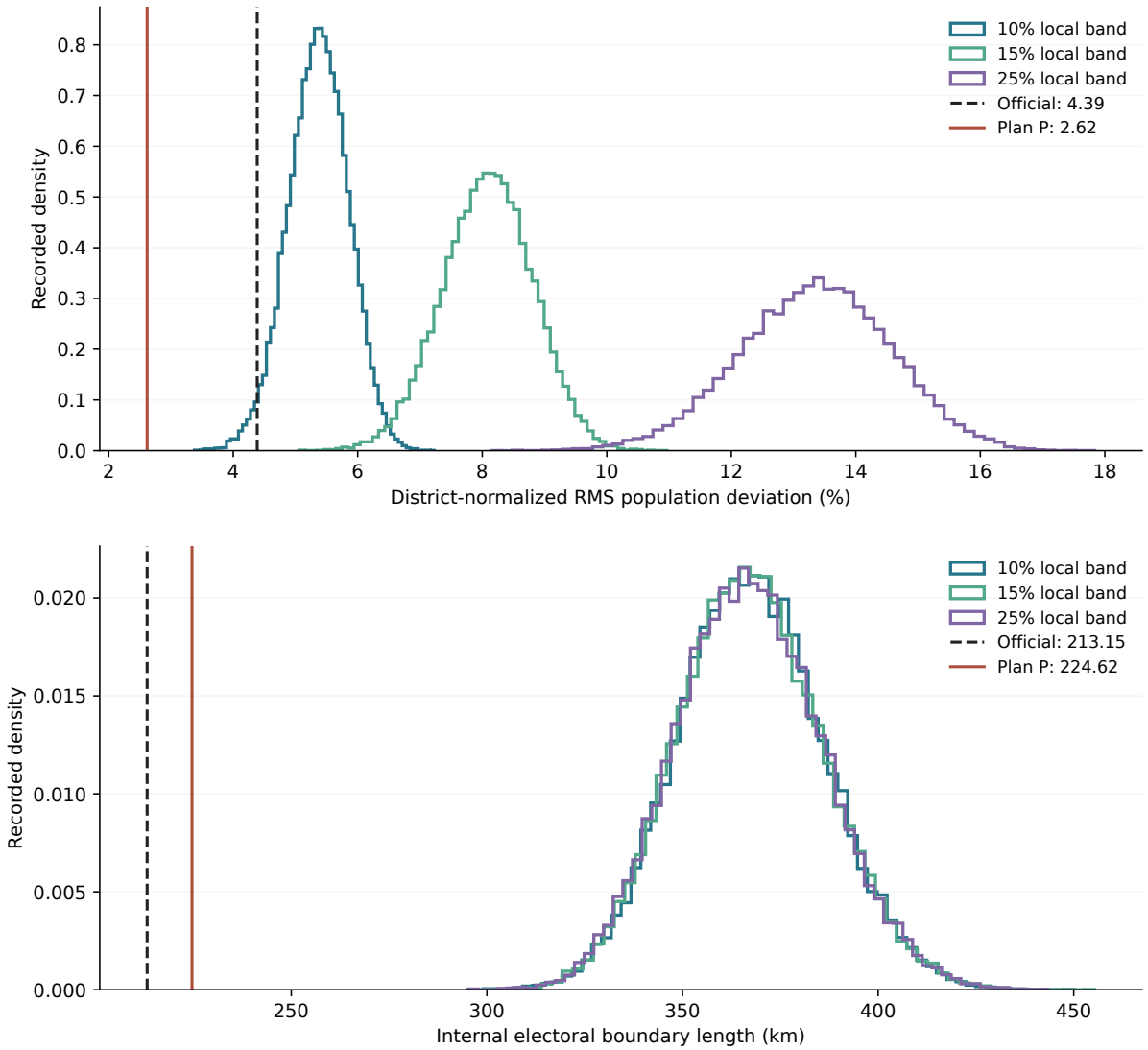


Figure 1: Monte Carlo distributions under three fixed local population bands. The marked official map and the population-oriented Plan P use the same atom graph and census counts as the records. These are conditional descriptive comparisons, not calibrated p-values. Plan F is evaluated separately in Section 5.

The endpoints are deliberately contrasting assignments. They do not establish how independents' supporters would have voted between different candidates. Nor do these five common allocations exhaust the possibilities of assigning a different fraction to each source area.

To compare a map at a given territory-wide support level x, use a uniform additive swing with clipping:

$$z_u(r,x) = \min\{1, \max\{0, a_u(r) + \delta(r,x)\}\}, \qquad \frac{\sum_u t_u z_u(r,x)}{\sum_u t_u} = x. \tag{4.2}$$

The scalar δ is found by bisection. The weighted mean is continuous and nondecreasing in δ and spans $[0,1]$, so a solution exists for every x in the interior. We retain observed valid-vote totals t_u as turnout weights; no population-to-vote conversion is performed.

For constituency c in map π, modeled group A support is

$$z_c(\pi; r, x) = \frac{\sum_{u \in V_c(\pi)} t_u z_u(r,x)}{\sum_{u \in V_c(\pi)} t_u}. \tag{4.3}$$

The design grid uses $x = .40, .425, .45, .475, .50, .525, .55, .575, .60$. This is a set of historical counterfactuals at nine support levels. It is not a forecast of support or turnout in a current Hong Kong election. Calibration to a common territory-wide vote share also does not imply equal votes within every administrative district.

4.3 A two-seat, single-vote election model

Suppose each modeled group fields two candidates. Within each group, a fraction q of its supporters votes for the stronger candidate and $1-q$ for the other, with

$$q \in \{.50, .55, .60, .65, .70\}. \tag{4.4}$$

For constituency support z, the four candidate vote shares are qz, $(1-q)z$, $q(1-z)$, and $(1-q)(1-z)$. The top two candidates are returned. With a common split parameter for both groups, group A's seat count is

$$s_q(z) = 1 + \mathbf{1}\{z > q\} - \mathbf{1}\{z < 1-q\}, \tag{4.5}$$

away from ties. At $z = q$ or $z = 1-q$, tied candidates receive the symmetric expected seat count; thus the transition values are 1.5 and 0.5 respectively when $q > 1/2$. At $q = z = 1/2$, the expected count is one. Total modeled seats are

$$S_\pi(r, q, x) = \sum_{c \in \mathcal{C}} s_q(z_c(\pi; r, x)). \tag{4.6}$$

Every scenario returns 88 seats in total. The model is deliberately transparent about coordination: $q = 1/2$ represents equal splitting, whereas a larger q represents greater concentration on one candidate. A change in q can change seat totals without changing any boundary. Real candidacies, personal vote, nomination, and strategic coordination are not identified from our historical inputs.

4.4 The equal-support certificate and curve diagnostic

Define equal-support bias, in seat units, by

$$B_\pi(r, q) = S_\pi(r, q, .50) - 44. \tag{4.7}$$

The comparison asks whether the two modeled groups receive equal seats when their overall modeled support is equal. It does not require proportionality at every vote share, a normative condition that the two-seat rule itself need not deliver.

Definition 4.1 (Finite-model gerrymandering-free graph partition)**.** *Let* $\mathcal{R}_0 = \{0, .50, 1\}$ *and* $\mathcal{Q}_0 = \{.50, .60, .70\}$ *be the original nine-scenario family. A graph partition is* gerrymandering-free under the nine-scenario equal-support criterion *if it is feasible under the stated graph and population constraints and*

$$B_\pi(r, q) = 0 \quad \textit{for all } (r, q) \in \mathcal{R}_0 \times \mathcal{Q}_0. \tag{4.8}$$

This is a finite-model certificate, not a finding that political manipulation or electoral advantage is impossible outside those scenarios. We separately report the stricter 25-scenario equal-support test on the expanded allocation and splitting grids. Passing the nine-scenario criterion does not imply passing that stronger test.

For the broader seat–vote curve, define the reflection defect

$$D_\pi(r, q, x) = \frac{S_\pi(r, q, x) + S_\pi(r, q, 1 - x) - 88}{2}, \qquad \Delta_\pi = \max_{r,q,x} |D_\pi(r, q, x)|. \tag{4.9}$$

At $x = .50$, $D = B$. At other support levels, it measures the difference between the curve and its own symmetrized counterpart. We report Δ across all $5 \times 5 \times 9 = 225$ design-grid cases, even when the equal-support criterion passes. It is a descriptive symmetry statistic, not a causal estimate of seats gained through intentional manipulation.

Proposition 4.2 (Group-label invariance)**.** *Exchanging group A and group B, replacing* r *by* $1 - r$*, and replacing support* x *by* $1 - x$ *changes the signs of the matched bias statistics but leaves their maximum absolute values unchanged.*

Proof. Before swing, the exchanged support fraction is $1 - a_u(r)$. The clipping identity $\text{clip}(1 - a - \delta) = 1 - \text{clip}(a + \delta)$ shows that the calibrated exchanged profile at $1 - x$ complements the original profile at x. Constituency aggregation preserves complementation. Equation (4.5), with symmetric tie handling, satisfies $s_q(1 - z) = 2 - s_q(z)$. Hence matched total seats sum to 88, and the sign of each bias reverses. The parameter grids are closed under the required reflection, so maxima of absolute values are unchanged. □

The same invariance holds for the final design objective. It prevents the objective from naming a preferred group. It does not establish that the endorsement classification, candidate model, or attainable political options are sufficient descriptions of real elections. Invariance of a rule and symmetry of its resulting seat distribution are separate properties; our certificate checks the latter on its stated finite family.

4.5 A falsifiable designation

A claim of "gerrymandering-free" must identify the test it passed. A zero value at $x = .50$ says nothing by itself about the rest of the curve, another election's support pattern, asymmetric candidate splitting, alternative political groups, or source areas subdivided below the DCCA level. A finite grid can also miss changes between adjacent parameter values because seat functions have thresholds.

We therefore supplement the design grid with a denser post-design sensitivity calculation and preserve its complete results. This is a stress test, not an independent election dataset. Its role is to expose model dependence, including failures, rather than to provide a second opportunity to choose an attractive reported percentile. The objective is a defensible proposal with a verifiable conditional property, not a map labelled neutral merely because a fairness metric was omitted.

5 Plan F: design, comparisons, and maps

5.1 A symmetric design objective

Plan F is a deterministic boundary proposal rather than a draw from the uniform Monte Carlo law. We keep the reference ensemble unchanged when designing it. Optimized maps answer a design question; inserting them into the reference as though they were uniform samples would change the comparison distribution.

The design minimizes a lexicographic objective whose first components are

$$\left(\max_{r,q} |B_\pi(r,q)|, \quad \Delta_\pi, \quad \sum_{r,q,x} |D_\pi(r,q,x)| \right). \tag{5.1}$$

These prioritize equal-support neutrality, then the worst reflection defect on the whole design grid, then the accumulated defects. Subsequent components minimize squared excess beyond the aspirational local 5% population band, population reassignment relative to the official assignment, and local population mean-square deviation. All components are recorded so that the political-symmetry objective cannot be hidden inside a population-only score.

The search preserves a hard 10% local population bound and requires every constituency's induced subgraph to remain connected. It also retains the entire Tsing Yi source-area group S22–S31, rural Sai Kung Q01–Q05, and the previously flagged Hong King source area Q16 in their official constituencies. These are explicit design safeguards; they do not appear in the broad reference ensemble. Their inclusion therefore conditions the proposal more narrowly than the reference law.

Candidate moves transfer a single boundary atom, exchange two boundary atoms, or transfer two adjacent boundary atoms together. A transfer must touch the receiving constituency through an existing graph edge. Compensating moves in two or three different administrative districts are considered when individual district moves cannot improve the discrete symmetry objective. These joint moves can cross a score plateau while leaving the intermediate one-district feasibility checks intact.

We compare completed searches from different documented initial designs. The delivered assignment is selected by the declared objective, with deterministic ordering of candidate actions. The numerical search establishes the properties of the output, not global optimality or a proof that every possible feasible partition was considered. Vote data enter the objective to reduce measured asymmetry; the design is *group-label invariant*, rather than vote-blind. Confusing those descriptions would conceal how the proposed map was actually chosen.

5.2 Population, continuity, and boundary complexity

Plan F assigns all 452 source areas once, returns 44 nonempty connected constituency graphs, retains all 18 administrative districts and their constituency counts, conserves the census popu-

lation, preserves the stated safeguards, and meets the hard local 10% band. A separate audit checks these properties from the delivered assignment file rather than accepting the optimizer's success flags.

Relative to the official map, it reassigns 76 whole source areas containing 1,263,289 residents, or 17.044% of the total population. Local population RMS changes from 4.385% to 4.221%, and territory-wide CV from 13.133% to 13.119%. There are 12 constituencies outside the aspirational local 5% band. Internal cut length is 317.76 kilometres, compared with 213.15 kilometres in the official map.

These quantities expose real design costs. Population-oriented Plan P, which did not use electoral data, moved only 11 atoms covering 2.154% of residents and achieved local RMS 2.617%. Plan F trades some of that population and continuity performance for a stronger stated political-symmetry property. A map may therefore be preferable for one objective without dominating all alternatives. The numerical table, complete transfer list, and geographical figures make that choice reviewable.

Table 1: Official map and two complete designed alternatives. Symmetry statistics use the final 225-point electoral grid; population values use the same census counts.

Statistic	Official	Plan P	Plan F
Local population RMS (%)	4.385	2.617	4.221
Global population CV (%)	13.133	12.644	13.119
Internal cut length (km)	213.149	224.618	317.763
Population reassigned (%)	0.000	2.154	17.044
Maximum equal-support bias (seats)	7.000	8.000	1.000
Maximum reflection defect (seats)	7.000	8.000	1.000

5.3 Political-symmetry results

Plan F passes Equation (4.8): all nine original allocation–splitting scenarios return exactly 44 seats for each modeled group at equal overall support. It is therefore gerrymandering-free under that explicitly scoped nine-scenario equal-support criterion. Across the full 225-point design grid, its maximum absolute reflection defect is 1 seat, compared with 7 for the official map and 8 for Plan P. On the expanded 25-scenario equal-support test, 23 cases return exactly 44 seats and the maximum bias is 1 seat. The stronger all-25 zero-bias requirement is therefore not satisfied. A rooted mixed-integer feasibility search reached its 150-second solver limit without a feasible zero-bias incumbent; this does not prove that a passing map is impossible.

The reference comparison applies identical support and candidate models to every recorded map. Figure 3 displays the distribution of the maximum reflection defect. We report reference positions as conditional descriptive comparisons and retain the chain diagnostics for the electoral statistics. For the maximum reflection defect, the three reference means are 6.48, 6.45, and 6.48 seats; each central 90% recorded range is 4–10 seats. The largest seat-count split-$\widehat{R}$ across all 225 electoral cases and three bands is 1.0056, while the smallest estimated bulk ESS is approximately 1,278. The maximum-defect statistic itself has bulk ESS between about 7,172 and 14,709. These are diagnostics for specified summaries, not proof of full graph-space exploration. Neither an optimized proposal's low rank nor a large nominal sample count supplies a calibrated p-value or proves why the official map was drawn.

The sign of the equal-support bias can change when the allocation of unclassified votes or

the candidate split changes. Accordingly, the paper does not identify one political group as the beneficiary of the official map. It asks whether a transparent design can reduce symmetry defects under a declared family of historical scenarios, while reporting the model dependence of that answer.

5.4 Sensitivity outside the design grid

The post-design audit doubles the parameter-grid resolution: nine unclassified allocations, nine candidate splits, and 17 support levels produce 1,377 cases. Plan F has exactly zero equal-support bias in 56 of 81 allocation–splitting cases, with a maximum equal-support bias of 4 seats. Its largest curve reflection defect on that grid is 4 seats. These results limit the scope of the finite certificate and demonstrate that a zero-bias result at selected parameter values is not a universal political guarantee.

This additional calculation is deliberately retained even if it weakens the most attractive description of the proposal. It uses the same historical data, so it is not an independent electoral replication. Its value is to detect numerical-grid dependence. Further validation would require alternative elections, independently justified support-group classifications, and richer candidate and turnout models.

A conditional certificate cannot be promoted into an unrestricted label by omitting those checks. The map is a concrete proposal for further review, with a mathematical equal-support test that can be rerun or rejected under another model. The phrase "gerrymandering-free under the equal-support criterion" must travel with its scope whenever the result is described.

5.5 The delivered geography and graph

Figure 4 compares the complete proposal with the official assignment. Colours identify constituency suffixes within each district; the same colour across two administrative districts does not join those areas into one constituency. Pink marks source areas whose assignment changes. District letters, the transfer table, the complete assignment CSV, and the GeoJSON collectively specify the proposal; the image alone is not a statutory boundary description.

Figure 5 shows an adjacency-graph example in Sai Kung. Vertices are source areas, edges are authenticated shared polygon boundaries, vertex size reflects population, and vertex colour records constituency membership. The diagram displays a graph partition of exactly the same assignment used in the numerical audit. Its straight drawn edges are schematic connections and must not be read as travel routes.

Figure 6 provides larger land-only details for the densely divided urban areas. Constituency codes identify the proposed regions; highlighted source-area boundaries can be matched to the complete transfer table. The generalized land mask is cartographic context and does not change the electoral polygons or graph used in any computation.

Table 2: All whole-DCCA transfers in Plan F. Unlisted units retain their official assignment.

Unit	Name	Population	Official	Plan F
A06	Kwun Lung	15,570	A2	A1
A12	Sheung Wan	14,830	A1	A2
C02	Tai Koo Shing East	18,008	C1	C2

Unit	Name	Population	Official	Plan F
C04	Sai Wan Ho	18,771	C2	C1
C05	Aldrich Bay	17,925	C2	C1
C16	Braemar Hill	16,171	C1	C2
C20	Fort Street	14,489	C1	C2
C27	Kornhill Garden	13,083	C2	C1
D01	Aberdeen	19,459	D2	D1
D02	Ap Lei Chau Estate	12,089	D1	D2
D07	South Horizons West	14,947	D1	D2
D14	Shek Yue	17,621	D2	D1
D15	Wong Chuk Hang	16,316	D1	D2
E05	Charming	16,172	E1	E2
E07	Fu Pak	18,266	E2	E1
E08	Olympic	19,373	E2	E1
E16	Yau Ma Tei North	11,894	E1	E2
F06	Nam Cheong South	19,024	F2	F1
F08	Nam Cheong West	18,936	F2	F1
F09	Fu Cheong	21,213	F1	F2
F18	Lai Chi Kok North	15,545	F1	F2
G01	Ma Tau Wai	20,185	G1	G2
G02	Sung Wong Toi	19,078	G1	G2
G04	Ma Tau Kok	14,397	G1	G2
G08	Kadoorie	17,422	G2	G1
G19	Whampoa East	18,287	G2	G1
G20	Whampoa West	20,847	G2	G1
G21	Hung Hom Bay	18,224	G2	G1
H02	Lung Ha	15,897	H1	H2
H05	Fung Tak	14,456	H1	H2
H06	Lung Sing	18,282	H1	H2
H07	San Po Kong	18,199	H2	H1
H16	Tsz Wan West	18,747	H2	H1
H19	Tsz Wan East	18,495	H2	H1
J06	Choi Tak	14,553	J4	J3
J16	Po Tat	17,104	J3	J2
J19	Lam Tin	18,666	J1	J2
J27	Laguna City	22,718	J2	J1
J30	Hiu Lai	16,691	J2	J4
J38	Upper Ngau Tau Kok Estate	16,007	J4	J2
K08	Discovery Park	17,325	K1	K2
K17	Shek Wai Kok	12,198	K2	K1
K18	Lei Muk Shue West	15,261	K2	K1
L13	Sam Shing	16,423	L1	L2
L15	Yuet Wu	12,513	L2	L1
L18	Butterfly	15,488	L2	L3
L20	Lok Tsui	15,051	L2	L3
L22	San King	12,747	L3	L2
L24	Tin King	14,017	L3	L2
L27	Siu Hong	14,401	L3	L1
M04	Yuen Lung	13,995	M1	M3
M07	Nam Ping	14,106	M1	M3
M09	Yuen Long Tung Tau	13,928	M1	M3
M10	Shap Pat Heung North	14,081	M1	M3
M15	Ha Tsuen	21,060	M3	M1

Unit	Name	Population	Official	Plan F
M16	Ping Shan Central	16,850	M3	M1
M17	Shing Yan	14,300	M3	M1
M19	Tin Yiu	12,939	M3	M4
M20	Yiu Yau	13,688	M3	M4
M21	Tsz Yau	13,670	M3	M4
M28	Yuet Yan	18,418	M4	M3
M29	Ching King	19,055	M4	M3
M30	Fu Yan	19,008	M4	M3
P02	Chung Ting	13,725	P2	P1
P03	Tai Po Central	13,139	P2	P1
P09	Wang Fuk	26,893	P1	P2
Q23	Hau Tak	17,611	Q3	Q1
Q28	Wan Po North	18,682	Q1	Q3
R03	Wo Che Estate	17,572	R1	R2
R13	Chui Tin	14,635	R3	R1
R20	Tai Wai	19,924	R1	R3
R25	Hoi Nam	14,343	R4	R2
R35	Tai Shui Hang	18,198	R2	R4
R39	Bik Woo	16,368	R2	R1
S07	Kwai Chung Estate North	17,538	S3	S2
S16	Wah Lai	16,182	S2	S3

6 Interpretation, limitations, and reproduction

6.1 What the proposed map establishes

The completed study models real Hong Kong constituency geography as graph partitions, generates fixed-target Monte Carlo ensembles, evaluates the official assignment against those alternatives, and delivers a complete replacement assignment. It adds explicit voting scenarios and a partisan-symmetry objective rather than implying that population equality alone establishes neutrality.

The official map is relatively well balanced in the uniform 10% population reference and has shorter internal electoral-polygon boundaries than every recorded alternative. Those findings cannot establish an absence of gerrymandering. The voting calculation asks a separate question and can reveal asymmetry under one scenario even when the same map performs well on geometry. Conversely, an asymmetry score can arise from spatial support patterns and an election rule without proving political intent.

Plan F's results are stated as properties of its delivered graph partition under declared inputs. The method treats group names symmetrically, exposes unclassified votes, respects the two-seat rule in its model, and checks an equal-support criterion directly. The score and certificate can be reproduced from the files. These are useful properties of a proposal; they are not an assertion that every future election would be unbiased or that a boundary authority has approved the plan.

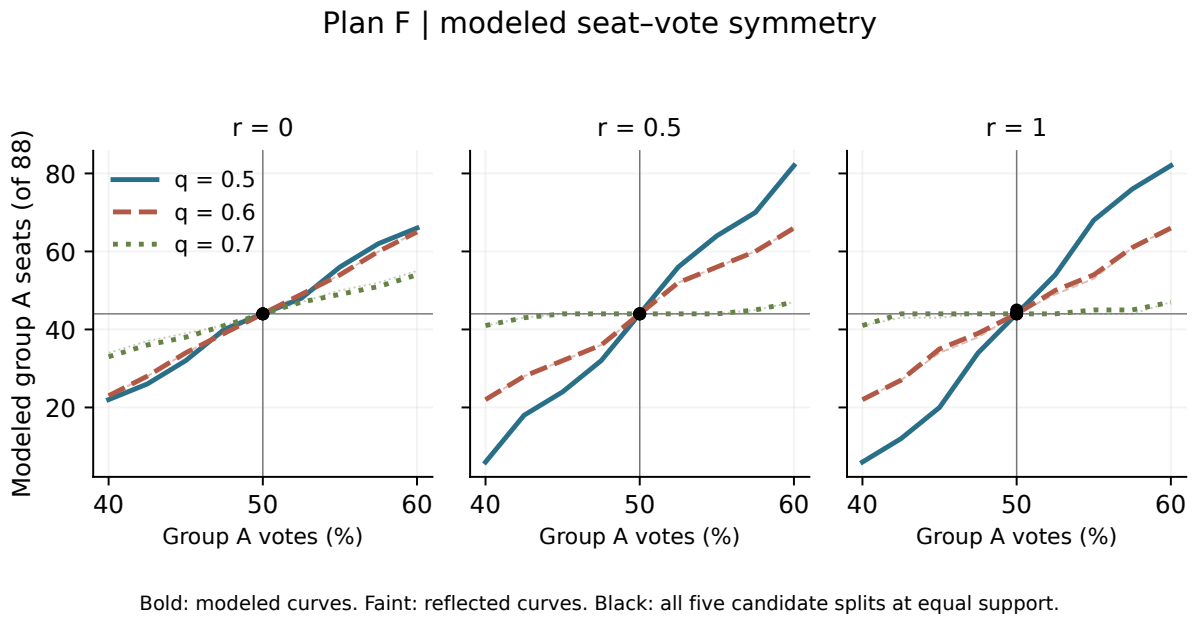


Figure 2: Seat–vote curves for Plan F under selected historical-support and candidate-splitting scenarios. Faint curves show reflected counterparts; coincidence indicates symmetry in the stated model. The full numerical certificate and all design-grid cases accompany the source archive. These curves are counterfactual calculations, not current election predictions.

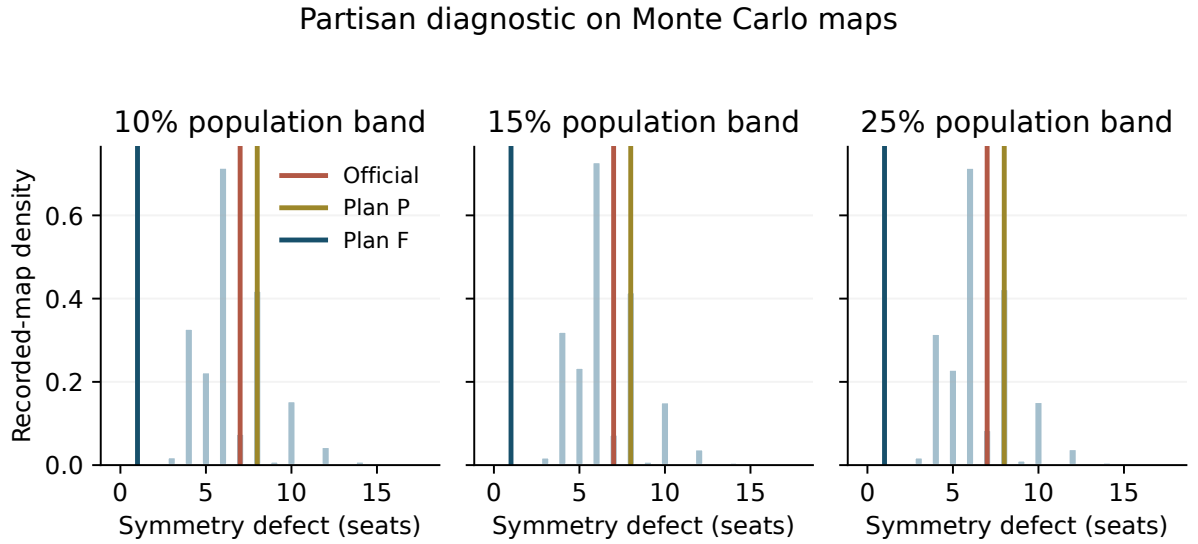


Figure 3: Maximum seat–vote reflection defects for the recorded Monte Carlo maps, with the official map and designed alternatives evaluated on identical electoral scenarios. Population-band changes define different reference spaces. Scores include geographical support patterns and voting-model assumptions; they do not measure intentional manipulation directly.

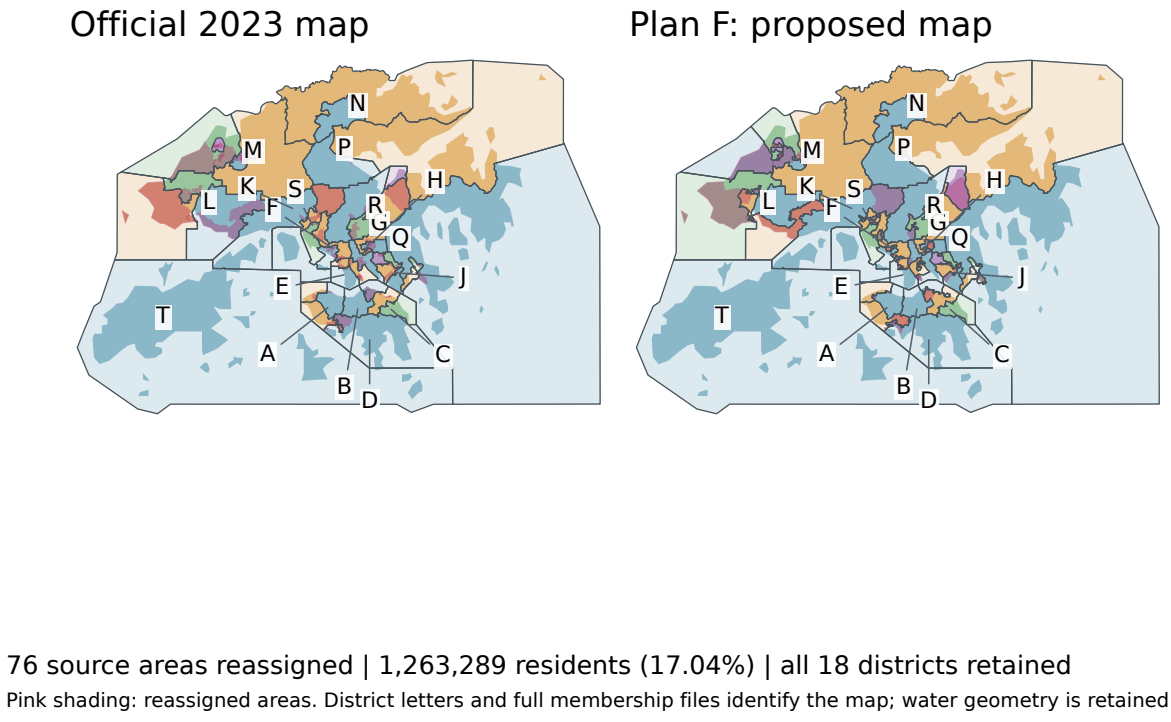


Figure 4: Official boundaries and the complete Plan F proposal. The electoral polygons retain marine areas; generalized Natural Earth land is cartographic context [15]. All source-area assignments, including areas too small to identify in this overview, are available in the reproduction files. A mathematical fairness certificate does not substitute for local community, transport, building, or legal review.

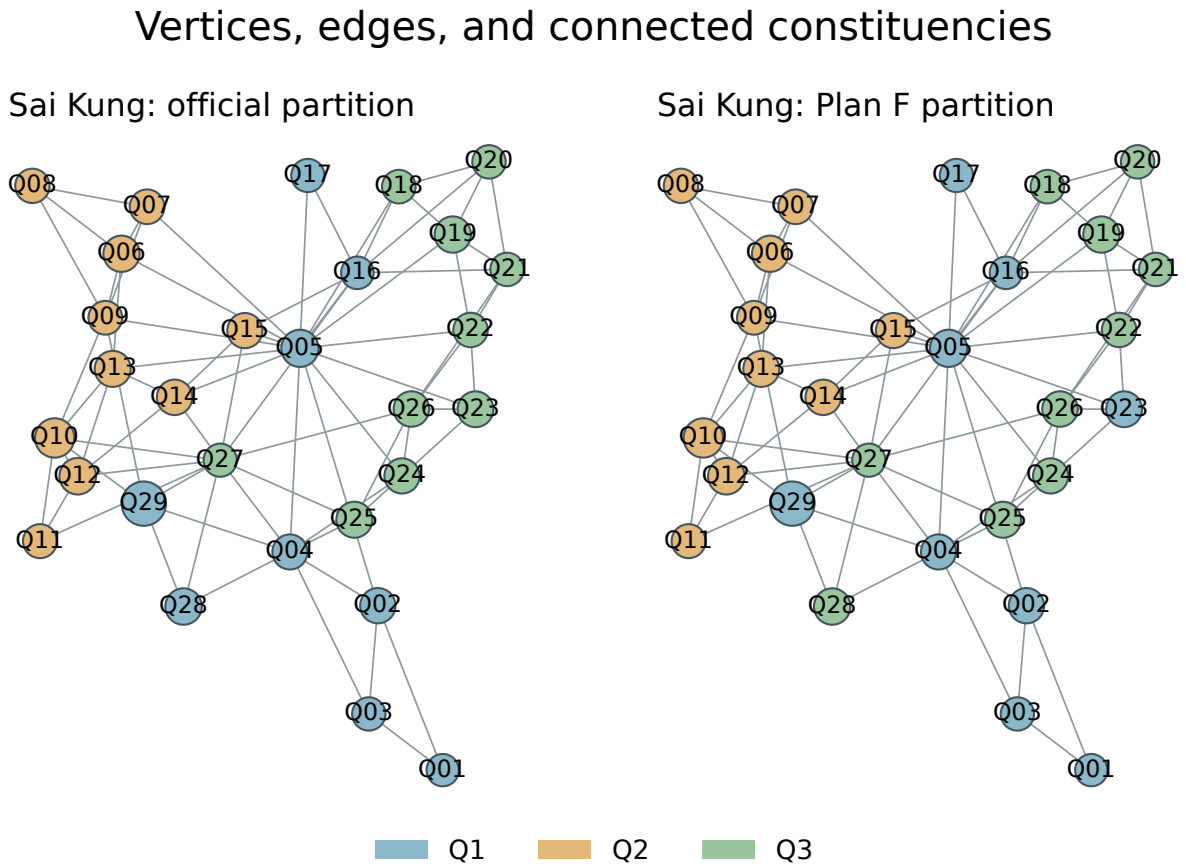


Figure 5: Constituencies represented as connected graph partitions in Sai Kung. The proposal's certificate concerns the territory-wide modeled seat comparison, not a separate guarantee for each administrative district. Graph edges measure polygon adjacency and do not certify physical access.

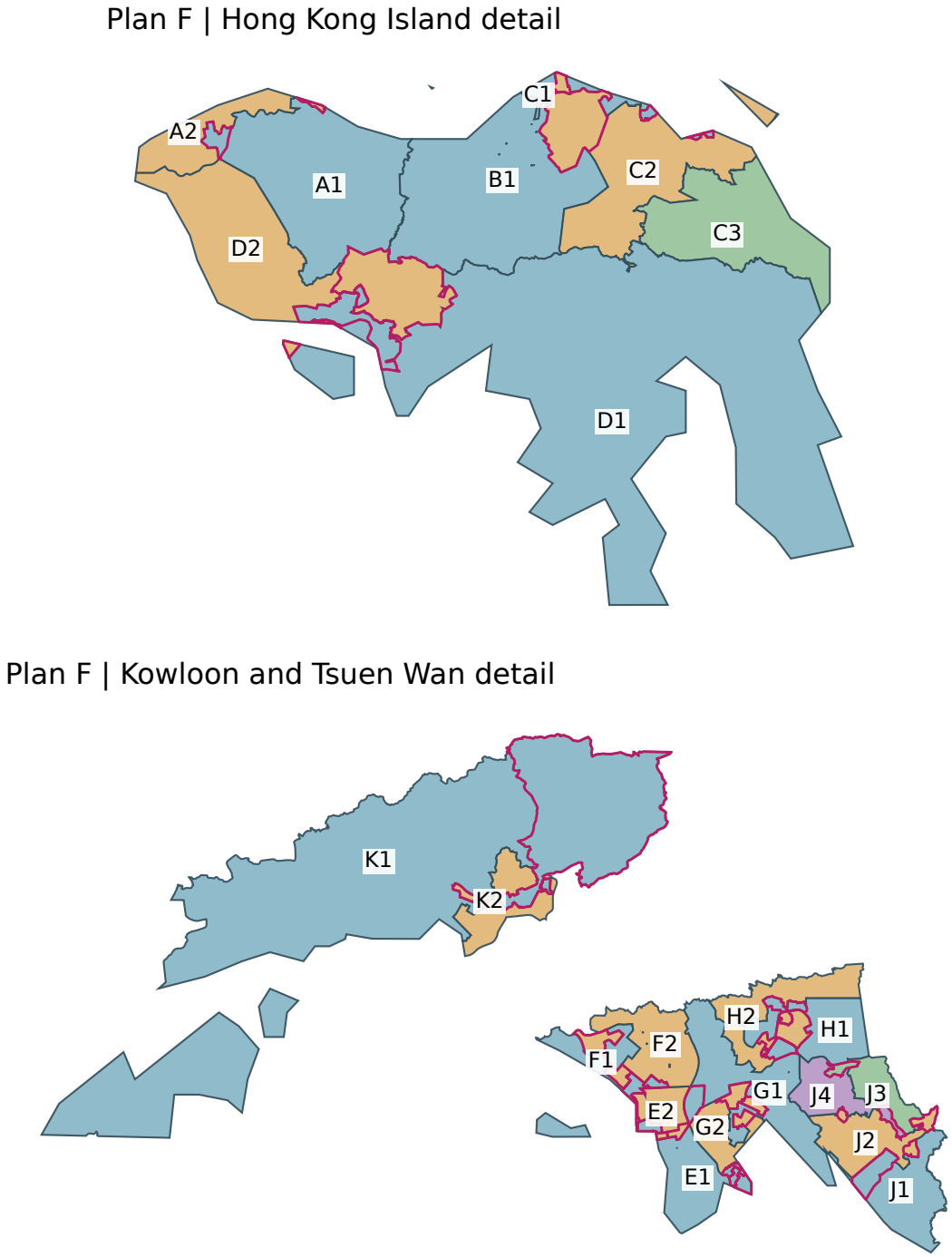


Figure 6: Urban details of Plan F. The letters and suffixes identify the complete proposed constituency assignment; pink outlines mark reassigned source areas. The full electoral polygons, including marine areas, are supplied in the GeoJSON.

6.2 Why redistricting alone is insufficient

The population decomposition places a hard limit on improvement with district nesting and allocation held fixed. Boundary moves can reduce the within-district variance, while the different district means remain. Removing that component would require changes to constituency or member allocation, or to the administrative nesting framework. The present proposal retains those institutional constraints, so it cannot be presented as a complete solution to territory-wide equality.

Electoral boundaries also cannot recreate an unobserved candidate choice set. Our two-group, four-candidate scenarios assume candidates are available and that supporters split between them in a specified way. Nomination and eligibility, numbers of candidates, personal support, coordination, and turnout can materially change outcomes without any boundary moving. Those mechanisms should be investigated independently rather than attributed to a graph partition score. The geographical seats are also only part of the District Council composition [11].

Consequently, the recommendation is a testable boundary proposal coupled with review of institutional allocation and electoral-model assumptions. Population fairness, partisan symmetry, local communities, and practical access are connected goals but remain distinct checks. No one number settles all of them.

6.3 Limits of the electoral evidence

The 2019 totals are observed votes for particular candidates under historical boundaries and a different seat arrangement. Their aggregation onto alternative boundaries does not identify how the same voters would respond to another candidate list. The endorsement grouping deliberately leaves approximately one-third of votes unclassified. Common allocation scenarios are a transparent sensitivity family, not proof of any unobserved individual's preferences.

The candidate-splitting model is symmetric by design. A real election may have different numbers of candidates in the two groups and unequal coordination parameters. A symmetry certificate under equal splitting rules cannot certify outcomes under those asymmetric mechanisms. Nor does a finite grid of support and splitting values cover every continuous parameter value; the denser sensitivity results are therefore substantive findings rather than optional qualifications.

The statistical reference is also conditional. Uniform connected partitions give complicated boundaries substantial collective weight. A community- or compactness-conditioned reference could yield different electoral and geometrical rankings. Its constraints would need independent justification, and its target and mixing properties would need fresh validation. We do not claim that the existing chain has explored every component of its feasible support.

6.4 Reproducible evidence and a shorter main paper

The accompanying source archive contains the input geometry and census crosswalk; the official result and ballot-body snapshots; the complete candidate grouping; all 96,000 recorded assignments; electoral scenario arrays; the proposed assignment and GeoJSON; search traces; numerical diagnostics; independent audits; figures; and the LaTeX source. Long topology and population audits, synthetic-grid state lists, original-manuscript corrections, allocation counterfactuals, and full trace plots are retained there rather than reproduced as a lengthy main-paper appendix.

The principal workflow is:

```
python redistricting/sample_ensembles.py
python redistricting/audit_ensembles.py
python redistricting/collect_votes.py
python redistricting/prepare_partisan.py
python redistricting/reproduce_fair_design.py
python redistricting/analyze_partisan.py
python redistricting/report_fair_plan.py
python redistricting/audit_fair_plan.py
tectonic main.tex
```

The snapshot files make the electoral extraction reviewable if a live website changes. The sample records allow the new electoral diagnostics to be regenerated without rerunning 108 million clock steps. Exact random trajectories require the recorded runtime and NumPy versions; numerical comparisons and the independent feasibility audit are also supplied in saved results.

The final proposal should be evaluated through its assignments, stated objective, fairness criterion, and documented costs. “Gerrymandering-free” is defensible here only as the explicitly scoped finite-model property. A broader political claim would require evidence and tests that the available historical data do not supply.

References

[1] Electoral Affairs Commission. *Report on the 2023 District Council Ordinary Election*, detailed report, Chapter 2, paragraphs 2.12–2.17. 2024.
https://www.eac.hk/pdf/distco/en/2023dc_report/2023dcereport_detailreport.pdf.

[2] Hong Kong Special Administrative Region. *District Councils (Amendment) Ordinance 2023*, Ordinance No. 19 of 2023, especially sections 9, 84, 98, and 100.
https://www.legco.gov.hk/yr2023/english/ord/2023ord019-e.pdf.

[3] Electoral Affairs Commission. *Boundary Maps and Descriptions of District Council Geographical Constituencies for the 2023 District Council Ordinary Election*. Eighteen English district-description PDFs, codes A–T excluding I and O. 2023.
https://www.eac.hk/en/elections/distco/2023dc_boundary/2023dc_elect_map.html. Individual URLs and hashes accompany the extracted data.

[4] Electoral Affairs Commission. "Provisional recommendations on boundaries and names of District Council Geographical Constituencies for 2027 District Council Ordinary Election for public consultation commences today." Government press release, August 25, 2026.
https://www.info.gov.hk/gia/general/202608/25/P2026082500262.htm.

[5] Electoral Affairs Commission. *District Council Geographical Constituency Boundaries of the 2023 District Council Ordinary Election*. Dataset metadata, DATA.GOV.HK.
https://data.gov.hk/en-data/dataset/eac-eacpsi01-csdi-dc-boundaries-2023.

[6] Electoral Affairs Commission. *2019 District Council Ordinary Election Constituency Boundaries*. Official 452-DCCA digital geometry, EPSG:2326.
https://data.gov.hk/en-data/dataset/eac-eacpsi01-dcca-boundaries-2019;
https://www.eac.hk/datagovhk/dc2019/DCCA2019.json.

[7] Census and Statistics Department. *2021 Population Census Statistics by District Council Constituency Area*. Bilingual CSV dataset and geography definitions. https://www.census2021.gov.hk/doc/DCCA_21C.zip;
https://www.census2021.gov.hk/doc/2021-statistics-classification-idds-en.pdf.

[8] Daryl DeFord, Moon Duchin, and Justin Solomon. "Recombination: A family of Markov chains for redistricting." *Harvard Data Science Review* 3(1), 2021. https://doi.org/10.1162/99608f92.eb30390f.

[9] Sarah Cannon, Moon Duchin, Dana Randall, and Parker Rule. *Spanning Trees and Redistricting: New Methods for Sampling and Validation*. arXiv:2210.01401, updated manuscript; published in *SIAM Review*, 2026. https://arxiv.org/abs/2210.01401; https://doi.org/10.1137/24M166944X.

[10] Aki Vehtari, Andrew Gelman, Daniel Simpson, Bob Carpenter, and Paul-Christian Bürkner. "Rank-normalization, folding, and localization: An improved $\widehat{R}$ for assessing convergence of MCMC." *Bayesian Analysis* 16(2):667–718, 2021. https://doi.org/10.1214/20-BA1221.

[11] Registration and Electoral Office. *2023 District Council Ordinary Election: Election Brief*. Composition, nomination, and the two-seat single-vote rule. https://www.elections.gov.hk/dc2023/eng/ebrief.html.

[12] Registration and Electoral Office. *2019 District Council Ordinary Election: All Hong Kong Results*. Official candidate-level results, including the published correction. Snapshot extracted October 12, 2026.
https://www.elections.gov.hk/dc2019/eng/results_hk.html.

[13] Registration and Electoral Office. *List of Names and Emblems of Prescribed Bodies and Persons Requested by Validly Nominated Candidates for Printing on Ballot Papers*. Eighteen district lists, 2019.
https://www.elections.gov.hk/dc2019/eng/names_and_emblems.html.

[14] Gregory Herschlag, Han Sung Kang, Justin Luo, Christy Vaughn Graves, Sachet Bangia, Robert Ravier, and Jonathan C. Mattingly. *Quantifying Gerrymandering in North Carolina*. 2018.
https://arxiv.org/abs/1801.03783.

[15] Natural Earth. *1:10 million physical vectors: land*, version 5.1.1. Generalized cartographic context only; public-domain data. https://www.naturalearthdata.com/downloads/10m-physical-vectors/10m-land/.